

Particulate Matter Monitoring & Health Effects

Issue 2026-08-14 | Entry window 14 August 2026 | 14 records carried, 17 rejected

14RECORDS CARRIED
(17 REJECTED)**9**SUBTOPIC CLUSTERS
REPRESENTED**11**EFFECT ESTIMATES
WITH INTERVALS**5**INSTRUMENTATION /
EXPOSURE RECORDS**0**METADATA-ONLY
(NO ABSTRACT)

Signal of the day

Two records landed on the same day that, read together, put a number on the exposure-misclassification ceiling — and it is low enough to matter.

Makacha et al. (*Front Glob Womens Health*) ran 5-day-per-season minute-resolved *personal* PM_{2.5} monitoring on 343 parous women in The Gambia, Kenya and Mozambique — 3,190 person-days — against reference monitors at nearby health facilities. Ambient-to-personal concordance was **0.01–0.56 with regression slopes below 1 in every setting**, and exposure was consistently higher *away* from home, peaking around 12:00–16:00. A fixed-site monitor is not a weak proxy for personal dose in these settings; across the worst site pairings it is barely a proxy at all.

Set that beside Atalell et al. (*Environ Res*), who detect developmental vulnerability at school entry in 54,458 Western Australian children at a mean PM_{2.5} of **7.40–7.54 µg/m³** — below the WHO annual guideline — with **OR 1.06 (1.03–1.08) for vulnerability in two or more domains per 1 µg/m³**. The effect size per unit is small, so the inference rests entirely on exposure contrast being measured cleanly. One study says effects are detectable at a 1 µg/m³ resolution; the other says the assignment method used to get there can be off by more than that. The gap between them is the case for dense, personal-scale, time-resolved measurement, and it is a measurement argument rather than an epidemiological one.

What changed today

- ▶ **A dependence-aware meta-analysis erases a cancer signal.** Negahi et al. pooled 25 estimates from 14 studies of climate-sensitive exposures and cancer. The conventional random-effects estimate was **1.07 (1.02–1.12)**; under robust variance estimation, accounting for multiple estimates drawn from shared cohorts, it became **1.02 (0.93–1.11)**. $I^2 = 97\%$, GRADE low or very low in all ten groups. This is a methodological result more than a substantive one, and it should be cited whenever a pooled air-pollution estimate is built from overlapping cohorts.
- ▶ **Downscaling needs bias correction before super-resolution.** d'Hondt et al. (preprint) argue that existing deep-learning downscalers implicitly treat the coarse CTM field as a blurred copy of the fine field, conflating cross-resolution *model* differences with genuinely missing spatial detail. Separating structural correction from spatial refinement is directly relevant to any LCS-plus-CTM fusion scheme, where the sensor network supplies exactly the detail the coarse model lacks.
- ▶ **A data-quality index that scores a low-cost sensor at 58/100.** Bajaj & Settle decompose fitness-for-use into accuracy, calibration, completeness, representational consistency and timeliness. On five July days of OpenAQ Los Angeles data a reference monitor scored ≈ 99 and an LCS ≈ 58 — penalised not mainly for accuracy but for completeness, outliers during a 4 July episode, and reporting that stopped altogether. The framing is worth borrowing even though the demonstration is $n = 2$.
- ▶ **Composition, not mass, again carries the toxicology.** Ayala et al. resolved 54 Catalan PM₁₀ filter samples into four MCR-ALS components and regressed six SH-SY5Y endpoints against them (R^2_{CV} 0.53–0.75). Winter biomass-burning markers (levoglucosan, dehydroabietic acid, PAHs) drove ROS, p53 activation and cell death; biogenic SOA tracers were unimportant, and a spring–summer SOA component inhibited acetylcholinesterase *without* cytotoxicity. Two distinct neurotoxic pathways from one mass metric.
- ▶ **An interventional record worth tracking rather than reading.** Wittkopp et al. published the protocol (NCT05994937) for a double-blind sham-controlled bedroom HEPA trial in adults with prediabetes, with continuous glucose monitoring as the primary outcome and continuous indoor PM_{2.5} throughout. No results yet; logged to `trials.json` for the weekly watch.

Corpus analytics

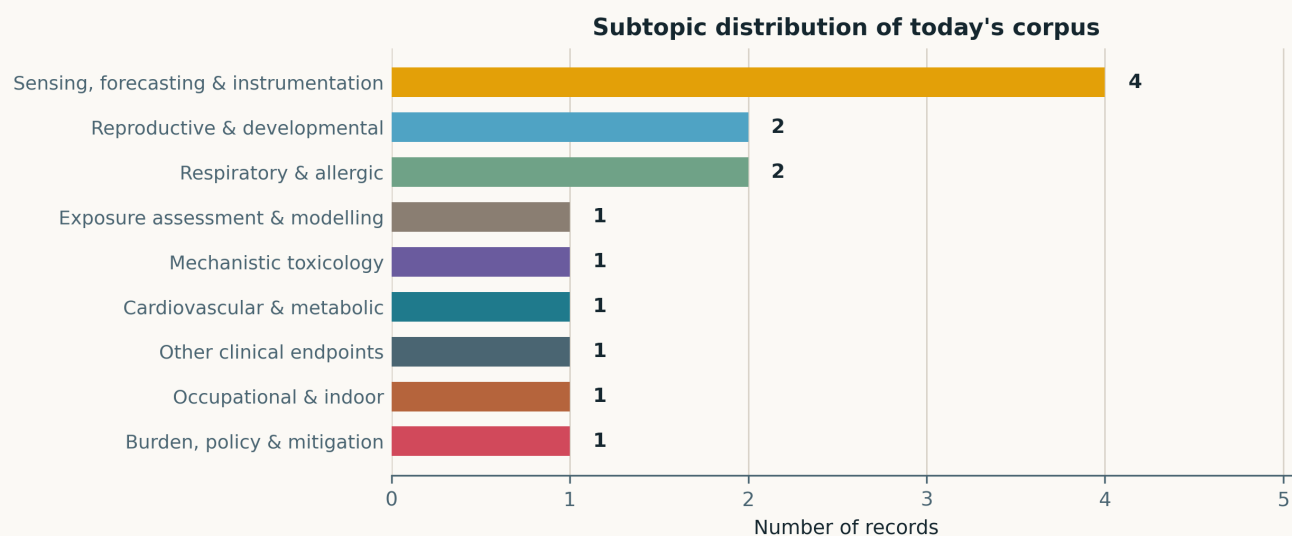


Fig. Subtopic assignment of the 14 carried records, single-label by dominant contribution. Nine of the ten clusters are occupied — the broadest spread in a week — but no cluster reaches five. *Sensing, forecasting & instrumentation* leads at 4; *Neuro / mental health* is the only empty cluster, which is unusual and is an artefact of the day rather than a trend.

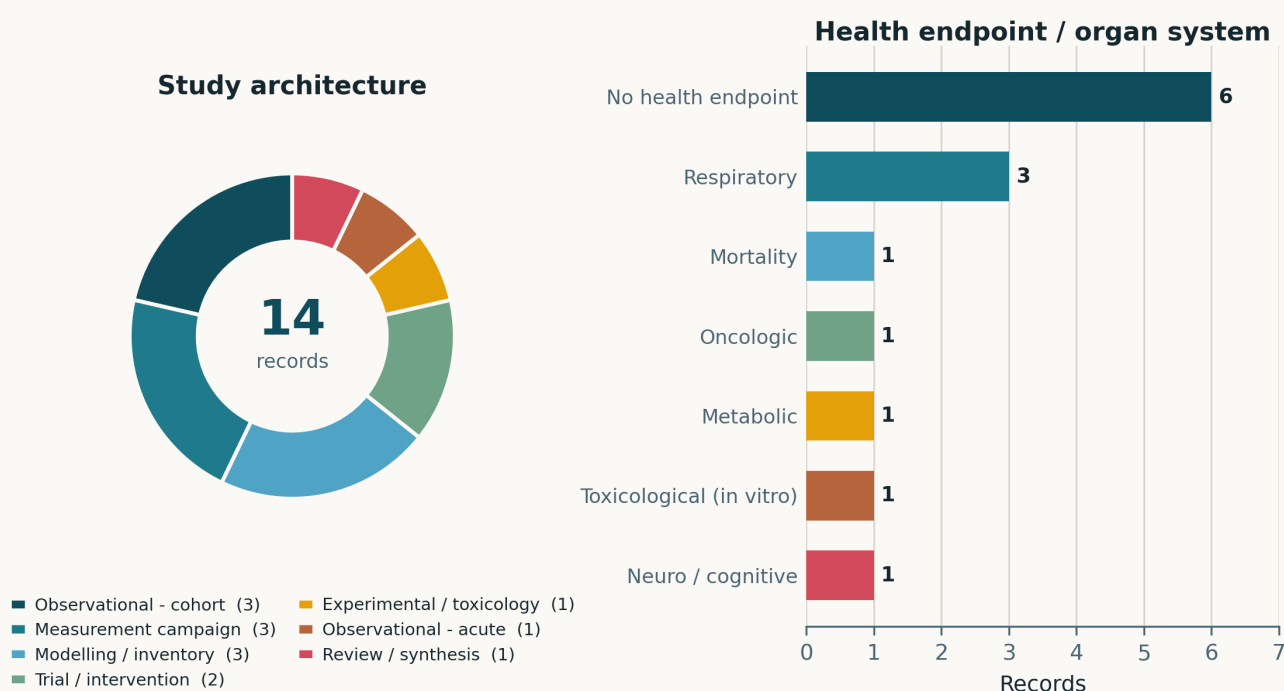


Fig. Left — study architecture. Observational designs (4), modelling (3) and measurement campaigns (3) split most of the corpus; the two records in the *Trial / intervention* column are a protocol and a feasibility pilot, so neither reports an outcome. Right — health endpoint. Respiratory endpoints lead at 3, and six records carry no human health endpoint — the four sensing records, the personal-exposure campaign, and the second-hand-smoke pilot, whose reported outcome is indoor $PM_{2.5}$ rather than a clinical measure.

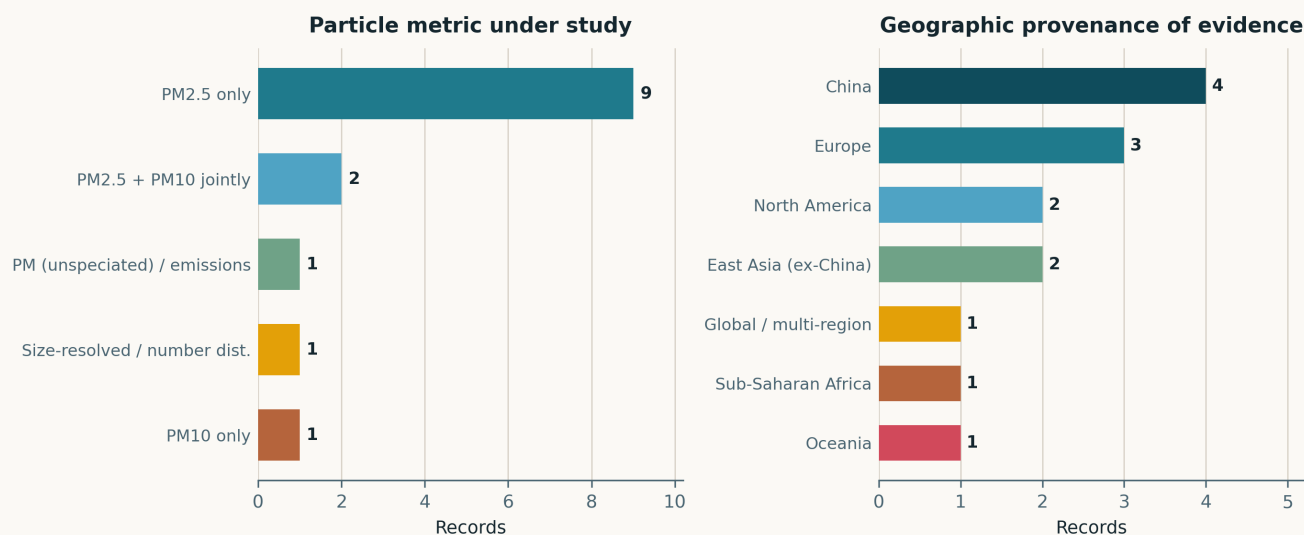


Fig. Left — particle metric. PM_{2.5} dominates, but two records depart from mass entirely: particle number size distribution and CCN in the Lhasa campaign, and a burden estimate carrying no PM metric at all (CO/NO₂/SO₂ only). Right — geographic provenance. China contributes four records and Europe three; sub-Saharan Africa is represented by a first-party personal-monitoring campaign rather than, as is more usual in this watch, a review of its absence.

Life-course windows addressed today (records may span several stages)

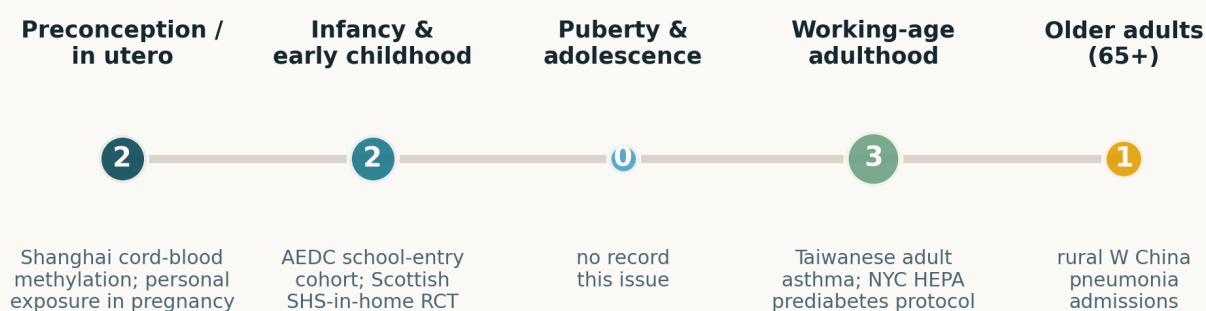


Fig. Life-course windows addressed. Counts are non-exclusive and describe this issue only. The in-utero window is occupied for the first time since 10 August (Shanghai cord blood; personal monitoring in pregnancy), and puberty/adolescence is empty — the mirror image of the 13 August distribution.

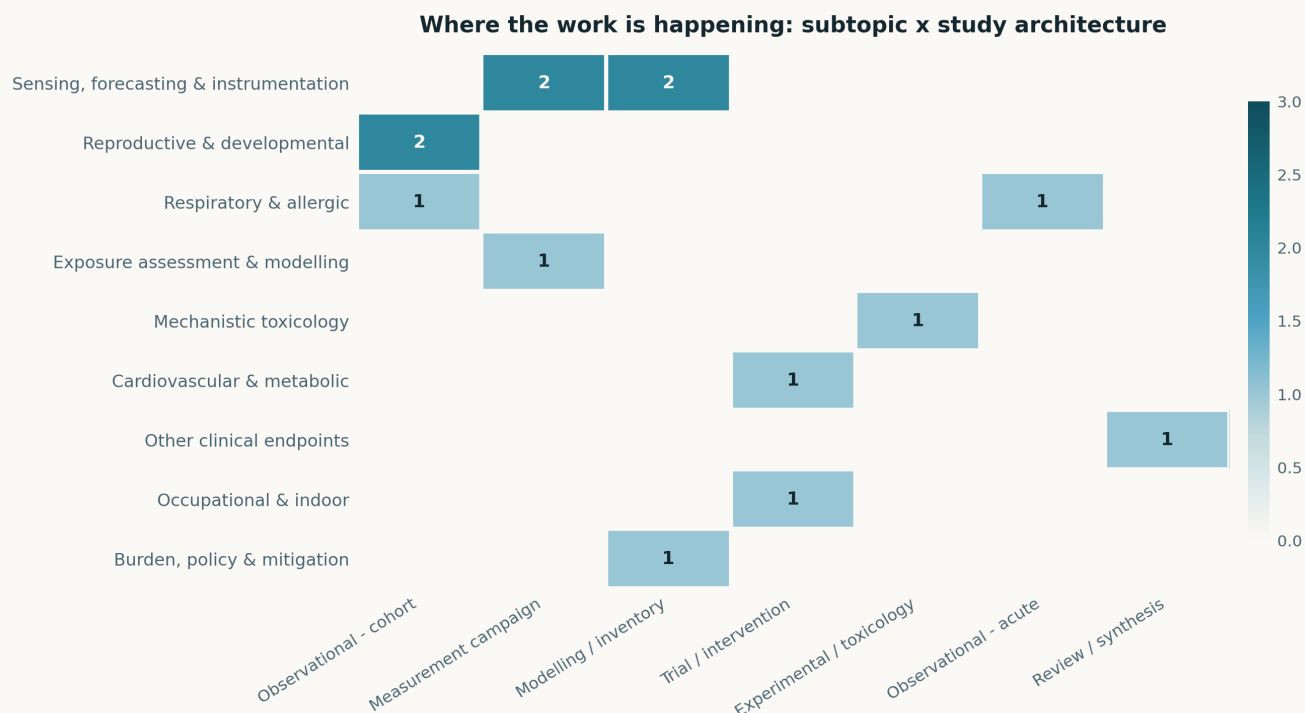


Fig. Subtopic $\times$ study-architecture cross-tabulation. The matrix is almost perfectly diagonal — nine subtopics spread across seven architecture columns with no column holding more than three records — which is the visual signature of a broad, shallow day. The instrumentation $\leftrightarrow$ epidemiology silo flagged on 12 and 13 August persists: every *Sensing* and *Exposure* record sits in a measurement or modelling column and none carries a health endpoint. Ayala et al. are the partial exception and the only record today in which exposure and outcome were measured by the same group on the same samples.

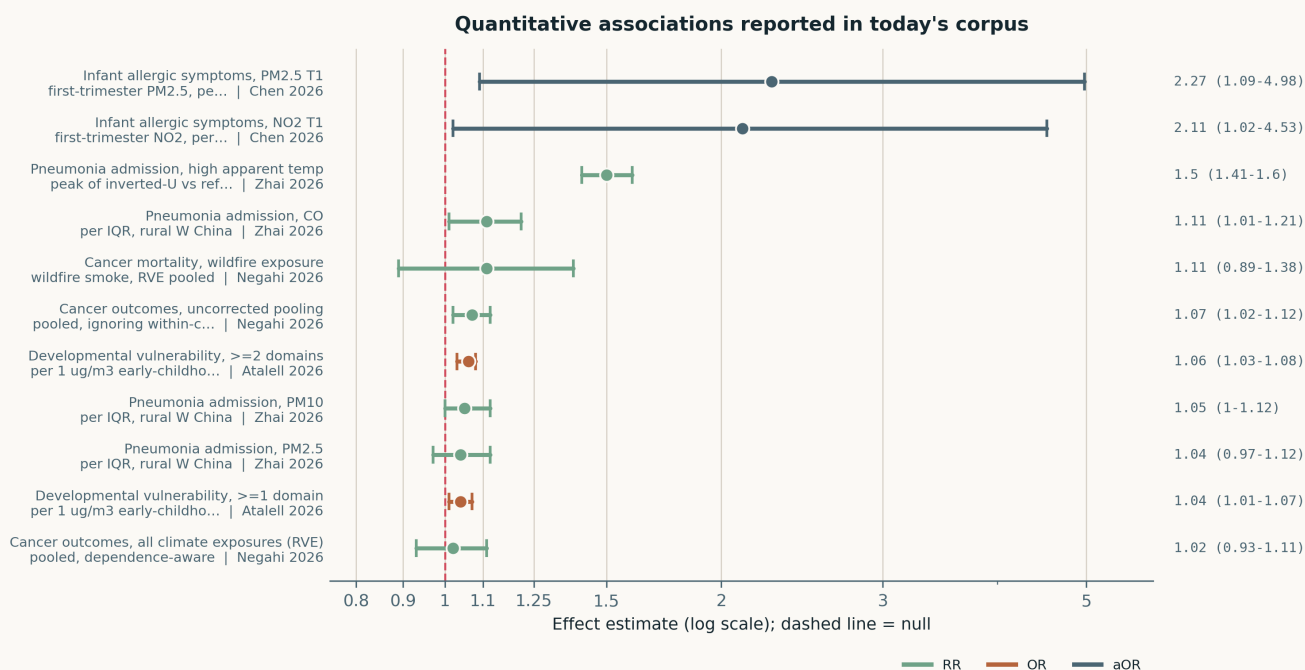


Fig. The 11 interval estimates carried today, on a common logarithmic axis. **The exposure contrasts are not comparable across sources.** Atalell et al. report per-1 $\mu\text{g}/\text{m}^3$ ORs from a whole-of-population cohort, which is why their intervals are the tightest on the plot and their point estimates the closest to unity; Chen et al. report per-unit adjusted ORs from 101 mother–infant pairs, which is why theirs are the widest; Zhai & Meng report per-IQR relative risks; and the three Negahi et al. estimates are pooled meta-analytic risk ratios, two of which are fitted to the same underlying studies under different dependence assumptions and are therefore not independent of one another. Makrufardi et al. report six odds ratios without intervals and are not plotted.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

4 records

d'Hondt et al. 2026 Research Square (preprint)

Neural downscaling of air-quality simulations requires structural correction before spatial refinement

Deep-learning downscaling of CAMS-class chemical transport model output, which is capped near 10 km for continental forecasts. The argument is that existing approaches either treat the coarse field as a smoothed copy of the fine field or learn a direct coarse-to-fine map, and both conflate two different things: cross-resolution differences in *model behaviour*, and genuinely missing spatial detail. Separating a structural correction step from the super-resolution step is what the paper proposes and tests. Directly transferable to LCS-plus-CTM fusion, where the network supplies the detail the model lacks and the model supplies the physics the network lacks — but a bias-corrected coarse field is the precondition. Preprint, not peer reviewed; carried on relevance, and no numeric claim from it is used in this issue's analytics.

doi: 10.21203/rs.3.rs-10627504/v1

Bajaj & Settle 2026 Environ Monit Assess

Rapid assessment of air pollution data quality using a composite index

A Data Quality Index aggregating five 0–100 dimensions — accuracy (30%), calibration (20%), completeness (20%), representational consistency (15%), timeliness (15%) — with weights the authors state should vary by stakeholder. Demonstrated on hourly OpenAQ PM records for Los Angeles, 1–5 July 2025: a reference-grade monitor scored ≈ 99 and a low-cost sensor ≈ 58 , the latter penalised for calibration bias, low completeness, extreme outliers during the 4 July episode, and zero timeliness once reporting ceased on 5 July. The value is in keeping the components legible rather than collapsing to a single opaque score. The demonstration is $n = 2$ over five days and includes a fireworks episode — an outlier regime chosen, apparently, for illustration — so the numbers are a worked example, not a benchmark.

doi: 10.1007/s10661-026-15780-6

Yaqoob et al. 2026 Sci Rep

Semiparametric hybrid air quality forecasting using Prophet ensemble and feature selection

Prophet models the trend and seasonality of daily $PM_{2.5}$; the residuals are then modelled by bagging, boosting, stacking and voting regressors, with Lasso, SHAP and recursive feature elimination applied to control the high-dimensional lag/interaction feature space. Tested on Seoul daily data and a high-dimensional simulated dataset against XGBoost, SVR and LSTM baselines; Prophet + boosting with Lasso and SHAP gave the lowest RMSE. The residual-modelling decomposition is a sensible way to keep an interpretable trend component, but the abstract reports no RMSE values, no forecast horizon and no comparison against a persistence baseline, which is the benchmark that actually discriminates in daily PM forecasting.

doi: 10.1038/s41598-026-64590-w

Zhang et al. 2026 Atmos Chem Phys

High frequency of urban new particle formation on the Tibetan Plateau

In-situ particle number size distribution measurements at Tibet University, Lhasa (3,650 m), across spring and autumn campaigns, with days classified NPF / Undefined / Non-Event from PNSD evolution plus size-segregated counts and growth rates. NPF event frequency was **86.2% in autumn and 62.1% in spring** — extremely high by urban standards — and estimated CCN was $(2.3 \pm 1.5) \times 10^3 \text{ cm}^{-3}$ on NPF days versus $(1.3 \pm 0.7) \times 10^3$ on non-event days. Autumn events were associated with greater sulfur-precursor availability; no dominant control was identifiable in spring, which the authors state plainly. Relevant here as a reminder that number-based instrumentation resolves a regime mass-based monitoring cannot see at all; the high-altitude, low-condensation-sink setting limits transfer to lowland urban networks.

doi: 10.5194/acp-26-11473-2026

Exposure assessment & modelling

1 record

Makacha et al. 2026 Front Glob Womens Health

Spatio-temporal and structural determinants of personal $PM_{2.5}$ exposure in three sub-Saharan African geographies

Five days per season of minute-resolved personal $PM_{2.5}$ monitoring on 343 parous women in The Gambia, Kenya and Mozambique, 3,190 person-days, aligned to microenvironment, fuel use, occupation, wealth and meteorology, with facility-based ambient monitors for comparison. Daily medians were modest — 10.4 (7.6–14.7), 8.1 (6.8–9.8) and 10.9 (9.3–13.5) $\mu\text{g}/\text{m}^3$ — but episodic spikes dominated, and WHO 24-h exceedance was strongly seasonal (The Gambia **55.9% dry vs 13.6% wet**). **Ambient–personal concordance was 0.01–0.56 with slopes <1 throughout**. Exposure was higher away from home with a 12:00–16:00 peak, so clean-fuel transition alone will not close the gap. The 5-day-per-season design is thin for characterising episodic exposure and the authors do not report how much of the between-person variance it captures.

doi: 10.3389/fgwh.2026.1843221

Reproductive & developmental

2 records

Atalell et al. 2026 Environ Res

Low-level $PM_{2.5}$ and developmental vulnerability at school entry, Western Australia

Whole-of-population linkage of the Australian Early Development Census (2009, 2012, 2015 waves) to midwives' notifications, birth registrations and hospital morbidity data; 54,458 children assigned daily $PM_{2.5}$ at Statistical Area Level 2 across prenatal, early-childhood and cumulative windows; Bayesian multilevel logistic regression against the 2009 10th-percentile domain cut-offs. Mean exposure **7.40–7.54 $\mu\text{g}/\text{m}^3$** , below WHO guidance. Early-childhood exposure: **OR 1.04 (1.01–1.07)** for vulnerability in one or more domains and **1.06 (1.03–1.08)** in two or more, per $1 \mu\text{g}/\text{m}^3$, with signals in physical health, social competence and language/cognitive domains. Prenatal and cumulative windows were null. The authors concede the critical limitation themselves: prenatal and early-childhood exposures are highly correlated, so the window attribution is not identified. Area-level assignment at SA2 resolution over a $1 \mu\text{g}/\text{m}^3$ contrast is the second limitation, and the one this watch cares about.

doi: 10.1016/j.envres.2026.125493

Chen et al. 2026 *Pediatr Allergy Immunol**Cord blood DNA methylation as a mediator between prenatal air pollution and infant allergic symptoms*

Shanghai prospective birth cohort, 101 mother–infant pairs with spatiotemporally modelled residential exposure, caregiver-reported cumulative allergic symptoms to 6 months, and targeted bisulfite sequencing of naïve CD4⁺ T cells in a subset of 30 cord-blood samples. First-trimester PM (aOR 2.27, 1.09–4.98) and NO₂ (2.11, 1.02–4.53) were associated with allergic symptoms; third-trimester sulfate, nitrate and ammonium components were positively associated; a CpG in the *PRKACB* promoter showed statistical evidence of mediation for the sulfate association, supported by functional assays. The mediation arm rests on 30 samples and the outcome is caregiver-reported at 6 months, which is the weakest available allergic phenotype. The authors label the findings exploratory, correctly. Carried for the component-resolved exposure design, not the effect sizes.

doi: 10.1111/pai.70424

Respiratory & allergic**2 records****Makrufardi et al. 2026** *World Allergy Organ J**Ambient relative humidity and particulate matter vs. asthma treatment step*

288 newly diagnosed adults with asthma followed at least 6 months under GINA, with treatment step — not symptoms or spirometry — as the outcome, and both daily means and short-term differences in RH, temperature and PM_{2.5} as exposures. A 1 µg/m³ increase in the 7-day PM_{2.5} difference was associated with 1.052-fold odds of step 4 treatment and 1.056-fold odds of medium-dose ICS + LABA; a 1 µg/m³ increase in the 30-day difference gave 1.205-fold odds of triple therapy or biologics. Associations were clearer in the cold season. **No confidence intervals are reported in the abstract**, which is why none of these appear in the forest plot. Using variability rather than level as the exposure is the interesting choice — it maps onto what a high-time-resolution sensor measures well and a monthly average does not — but with a single centre and 288 patients this is explicitly hypothesis-generating, as the authors say.

doi: 10.1016/j.waojou.2026.101444

Zhai & Meng 2026 *Int J Environ Health Res**Apparent temperature, air pollutants and pneumonia hospitalisation in rural Western China*

Distributed lag non-linear models on NRCMS surveillance data from Jiuquan, 2015–2020, with stratification and explicit pollutant × temperature interaction terms. Apparent temperature showed an inverted-U with a peak RR 1.50 (1.41–1.60). Per-IQR increases gave CO 1.11 (1.01–1.21) and PM₁₀ 1.05 (1.00–1.12) as the only intervals excluding or touching unity; PM_{2.5} 1.04 (0.97–1.12), SO₂, O₃ and NO₂ were null. Synergy was reported between low apparent temperature and high pollution, and among pollutants. Risks were higher in children under 7 and in autumn/winter. A rural setting is a genuine gap-filler — most Chinese DLNM work is urban — but with six correlated pollutants and an interaction screen the multiple-testing burden is not addressed, and the PM findings are the weakest of the set.

doi: 10.1080/09603123.2026.2718327

Mechanistic toxicology**1 record****Ayala et al. 2026** *Environ Pollut**Contrasting neurotoxic pathways triggered by PM₁₀ and organic molecular markers, Catalonia*

Fifty-four PM₁₀ filter samples collected through 2022 at a rural background site (Bellver de Cerdanya), a suburban site (Manlleu) and a suburban-industrial site (Mollet del Vallès), analysed by GC-MS for 30 organic tracers and tested on SH-SY5Y neuroblastoma cells across six endpoints (viability, ROS, AChE activity, antioxidant and xenobiotic response, p53). MCR-ALS on the joint chemical-biological matrix resolved four components: winter biomass burning (levoglucosan, dehydroabietic acid, PAHs) driving cytotoxicity, oxidative stress and xenobiotic response; a year-round traffic component; a spring–summer SOA component producing **selective AChE inhibition without cytotoxicity**; and a summer POA component. PLS models were significant for five of six endpoints (R^2_{CV} 0.53–0.75, best for ROS, p53 and cell death). The strength is that exposure and effect were measured on the same extracts; the limits are one year, three sites, one immortalised cell line and organic-solvent extracts that do not represent inhaled particle dosimetry.

doi: 10.1016/j.envpol.2026.128961

Cardiovascular & metabolic**1 record****Wittkopp et al. 2026** *Am Heart J**Cleaner Air for Lower Cardiometabolic Risk: HEPA filtration trial protocol in prediabetes*

Protocol for a double-blind, randomised, sham-controlled trial of one month of bedroom HEPA filtration in adults with prediabetes in the New York City area (NCT05994937), with continuous indoor PM_{2.5} measured throughout. Primary outcomes are continuous glucose monitoring metrics before and after; exploratory outcomes include insulin resistance, serum biomarkers and transcriptomics. Secondary analyses will model measured PM_{2.5} as a continuous independent variable, which is the analysis that will actually be informative — the intention-to-treat contrast in filtration trials is routinely diluted by baseline indoor concentrations already being low. One month is far too short for any glycated-haemoglobin endpoint, which is presumably why continuous glucose monitoring was chosen instead. No results; logged to `trials.json`.

doi: 10.1016/j.ahj.2026.107573

Occupational & indoor exposure**1 record****O'Donnell et al. 2026** *medRxiv* (preprint)*Nicotine replacement therapy to reduce children's second-hand smoke exposure at home*

Pilot randomised controlled trial in Scotland testing whether free NRT plus telephone-delivered support is a feasible route to a smoke-free home, with objective indoor PM_{2.5} as the exposure measure rather than self-reported smoking — which is the design decision that makes it relevant here. Feasibility, recruitment and retention are the outcomes; the trial is not powered for exposure reduction. Notable as one of very few interventions targeting the indoor source directly rather than filtering downstream of it. Preprint, not peer reviewed; no numeric claim from it is used in this issue's analytics.

doi: 10.64898/2026.08.13.26360348

Burden, policy & mitigation

1 record

Wu & Cui 2026 J Glob Health*Aircraft CO, NO₂ and SO₂ emissions and public health at Chinese airports*

Landing-and-take-off cycle data from the Civil Aviation Administration of China, concentrations at each airport from a Gaussian diffusion model, premature deaths from the Global Exposure Mortality Model applied to populations near airports. Attributable mortality was higher in men than women and concentrated in older age groups; CO accounted for the most respiratory deaths; the peak burden year was 2019. **No particulate metric is estimated at all**, which for an aviation source — where ultrafine particle number is the exposure of concern and is not proxied by CO — is a substantive omission rather than a scope choice. A Gaussian plume model at airport scale with no measurement validation is the second limitation. Carried at tier C for the burden framing and as a marker of where aviation-adjacent monitoring is not being done.

doi: 10.7189/jogh.16.04210

Other clinical endpoints

1 record

Negahi et al. 2026 Int J Biometeorol*Climate-sensitive exposures and cancer outcomes: systematic review and meta-analysis*

Five databases searched to May 2026 for observational studies of wildfire, ambient temperature, hurricane or multi-hazard disaster, precipitation anomaly, or ambient PM_{2.5} against cancer incidence, mortality, survival or recurrence; 25 estimates from 14 studies pooled within ten exposure-outcome groups, with robust variance estimation as the primary method for groups containing dependent estimates. **Uncorrected pooled 1.07 (1.02–1.12); RVE 1.02 (0.93–1.11)** — the conventional estimate was inflated by correlated effects, with $I^2 = 97\%$. Wildfire and cancer mortality **1.11 (0.89–1.38)**; ambient PM_{2.5} associated with poorer survival; an apparently protective temperature-survival association vanished under RVE and is best read as residual confounding. GRADE certainty was low in two groups and very low in the other eight. Fourteen studies across five exposure types and four outcome classes is too thin a base for group-specific inference, and the paper says so.

doi: 10.1007/s00484-026-03293-3

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Wittkopp et al.	Cleaner Air for Lower Cardiometabolic Risk: protocol for a double-blind, randomized, sham-controlled trial of HEPA filtration in adults with prediabetes	<i>Am Heart J</i>	Trial / intervention	PM _{2.5} (indoor, continuous)	USA
Reproductive & developmental					
Atalell et al.	Exposure to low-level fine particulate matter air pollution and developmental vulnerability at school entry: a population-based cohort study in Western Australia	<i>Environ Res</i>	Registry cohort + geospatial	PM _{2.5}	Australia
Chen et al.	Cord blood DNA methylation as a potential mediator between prenatal ambient air pollution and early infant allergic symptoms in a Shanghai birth cohort	<i>Pediatr Allergy Immunol</i>	Prospective birth cohort	PM _{2.5} , PM _{2.5} components (SO ₄ , NO ₃ , NH ₄), NO ₂	China
Respiratory & allergic					
Makrufardi et al.	Effect of ambient relative humidity and particulate matter on the levels of asthma therapy	<i>World Allergy Organ J</i>	Prospective cohort	PM _{2.5}	Taiwan
Zhai & Meng	Impacts of apparent temperature and air pollutants on daily pneumonia hospitalizations and their interaction in rural areas of Western China	<i>Int J Environ Health Res</i>	Distributed-lag time-series	PM _{2.5} , PM ₁₀	China
Mechanistic toxicology					
Ayala et al.	Contrasting neurotoxic pathways triggered by PM ₁₀ in relation with organic molecular markers in suburban and rural sites in Catalonia	<i>Environ Pollut</i>	Exposure assessment + in vitro	PM ₁₀	Spain
Occupational & indoor					
O'Donnell et al.	Use of nicotine replacement therapy to reduce children's exposure to second-hand smoke in the home: a pilot randomised controlled trial in Scotland	<i>medRxiv (preprint)</i>	RCT process evaluation	PM _{2.5} (indoor)	Scotland
Sensing, forecasting & instrumentation					
Bajaj & Settle	Rapid assessment of air pollution data quality using a composite index	<i>Environ Monit Assess</i>	Monitoring-record analysis	PM _{2.5}	USA
Yaqoob et al.	Semiparametric hybrid air quality forecasting using Prophet ensemble and feature selection	<i>Sci Rep</i>	Machine-learning model	PM _{2.5}	South Korea
Zhang et al.	High frequency of urban new particle formation on the Tibetan Plateau: quantifying formation rates, growth rates and CCN production	<i>Atmos Chem Phys</i>	Supersite observation	Particle number size distribution, CCN	China
d'Hondt et al.	Neural downscaling of air-quality simulations requires structural correction before spatial refinement	<i>Research Square (preprint)</i>	Deep-learning model	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃	Europe
Exposure assessment & modelling					
Makacha et al.	Unequal exposure: spatio-temporal and structural determinants of personal PM _{2.5} exposure in three sub-Saharan African geographies	<i>Front Glob Womens Health</i>	Observational exposure study	PM _{2.5}	The Gambia / Kenya / Mozambique
Burden, policy & mitigation					
Wu & Cui	Impact of aircraft CO, NO ₂ and SO ₂ emissions on public health in Chinese airports	<i>J Glob Health</i>	Burden estimation (GBD)	CO, NO ₂ , SO ₂ (no PM metric)	China
Other clinical endpoints					
Negahi et al.	Climate-sensitive exposures and cancer outcomes: a systematic review and meta-analysis	<i>Int J Biometeorol</i>	Systematic review (PRISMA)	PM _{2.5} , wildfire smoke	Global

Provenance and method

Window. PubMed [EDAT] 2026-08-14 to 2026-08-14, contiguous with the 13 August issue; no gap. Every record was checked against `state/seen.json` (pmid → lowercased doi → title signature) before summarising; no cross-source duplicate arose this issue.

Per-leg yield. PubMed connector, two separate queries on the health and instrumentation axes: 8 and 5 hits, 13 distinct, **9 carried**. Local harvester: `pubmed_health` 15 and `pubmed_sensing` 1, of which five records were outside the connector's two queries and **2 of those were carried** (Zhai & Meng; Wu & Cui) — the fourth consecutive issue in which the local leg has earned its keep; Europe PMC 3 (**2 carried**, both preprints); Crossref by ISSN 7 (**1 carried** — three were cloud-microphysics papers from *Atmos. Chem. Phys.*, one was an *Atmos. Pollut. Res.* editorial board listing, one an Elsevier metadata-only VOC record). arXiv returned 146 kB with no entry on or after 10 August. **OpenAlex HTTP 429, 12th consecutive** — the leg remains broken and is not worth further effort. **Consensus** was re-queried on the calibration and drift axis and returned three highly relevant papers, **all out of window** (Crossref created 2026-02-19, 2026-03-26 and 2026-06-15); this is the seventh consecutive issue with zero Consensus carry, and the pattern is now clear enough to record: Consensus ranks by relevance, not recency, and cannot serve a one-day entry window. ClinicalTrials.gov: no PM or air-pollution registration created or updated inside the window; `trials.json` unchanged, though NCT05994937 now has a published protocol (Wittkopp et al., this issue).

Rejections (17). Fourteen in-harvest: one published erratum; one *JAMA* Medical News item reporting a study already covered; three non-PM toxicology papers (chlorpyrifos, ingested polypropylene microplastics + DEHP, ethanol neuroinflammation); a greenhouse-gas inventory for source-separated organics; a fly-ash/farmyard-manure agronomy study; a scoping review whose literature search closed in April 2021; two cloud-microphysics papers and one ice-nucleating-protein paper; a Permian Basin hydrocarbon source-apportionment study with no PM metric and no abstract released; a journal editorial board listing; and an editorial commentary on emerging POPs. Three further records were in scope but outside the entry window (the Consensus hits above). All are logged with reasons to `state/rejected.jsonl`.

Two records are preprints (Research Square, medRxiv) and are labelled as such in the register. Neither supports a numeric claim in the analytics.

Labels. Subtopic, design, particle-metric and geography labels were drawn only from vocabularies already mapped in `plots.py` and `mktable.py`; no new leaf label was written this issue. Assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. `check_dois.py` returned 0 fail, 0 warn across all 14 records.

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